

Q1 (2020 T3 Midterm Exam Q1a)

For the circuit shown in Figure 1, calculate the equivalent resistance R_{eq} as seen from terminal a-b.

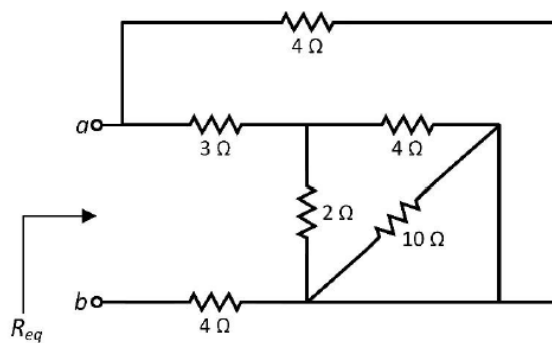


Figure 1

Answer: 6.08 Ω

Q2 (2021 T3 Midterm Exam Q1a)

An attenuator is an interface circuit that reduces the voltage or current level without changing the output resistance. For the circuit shown in Figure 2, calculate:

- Output current (i_o) in the circuit without attenuator as shown in Figure 2(a).
- Output current (i_o) in the circuit with attenuator as shown in Figure 2(b).
- Equivalent resistance (R_{eq}) of the circuit with attenuator in Figure 2(b) when the current source is turned off ($i_g = 0A$).

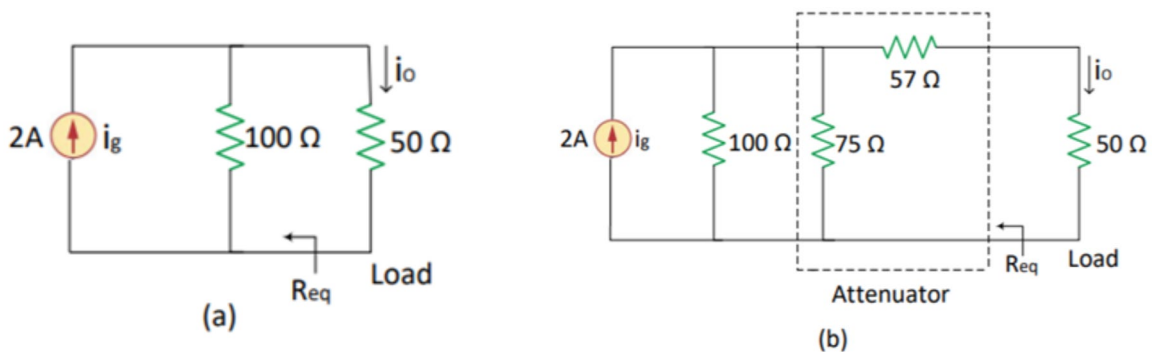


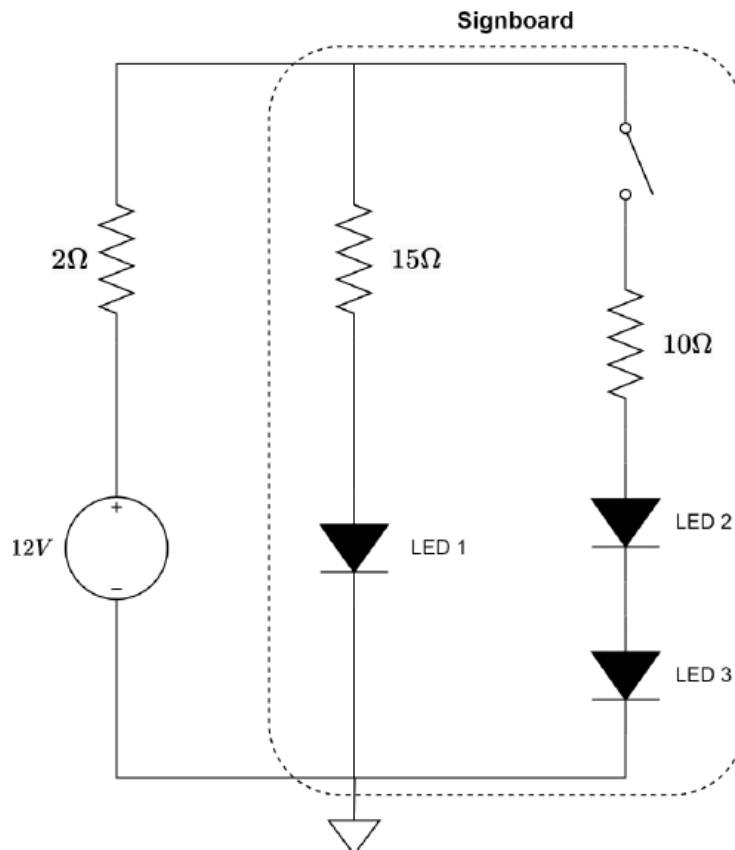
Figure 2

Answers: i) 1.333 A, ii) 0.572 A, iii) 99.86 Ω

Q3 (2023 T3 Midterm Exam Q2)

The circuit provided in the figure below regulates the brightness of the LEDs of an emergency exit signboard in two different scenarios: (a) during non-emergency situations and (b) during evacuation. The forward voltage of all LEDs is equal to 2.1 V.

- i) During non-emergency situations, the switch is **open**, and the signboard operates at a dim setting. Calculate the current flowing through the $15\ \Omega$ resistor and the power in LED 1 in this scenario.
- ii) In the event of an evacuation, the signboard operates at maximum brightness, with the switch **closed**. Determine the current through the $10\ \Omega$ resistor and the power in LEDs 2 and 3 in this scenario.



Answers: i) 0.582 A, 1.222 W, ii) 0.564 A, 1.184 W

Q4 (Modified from 2023 T1 Midterm Q2a)

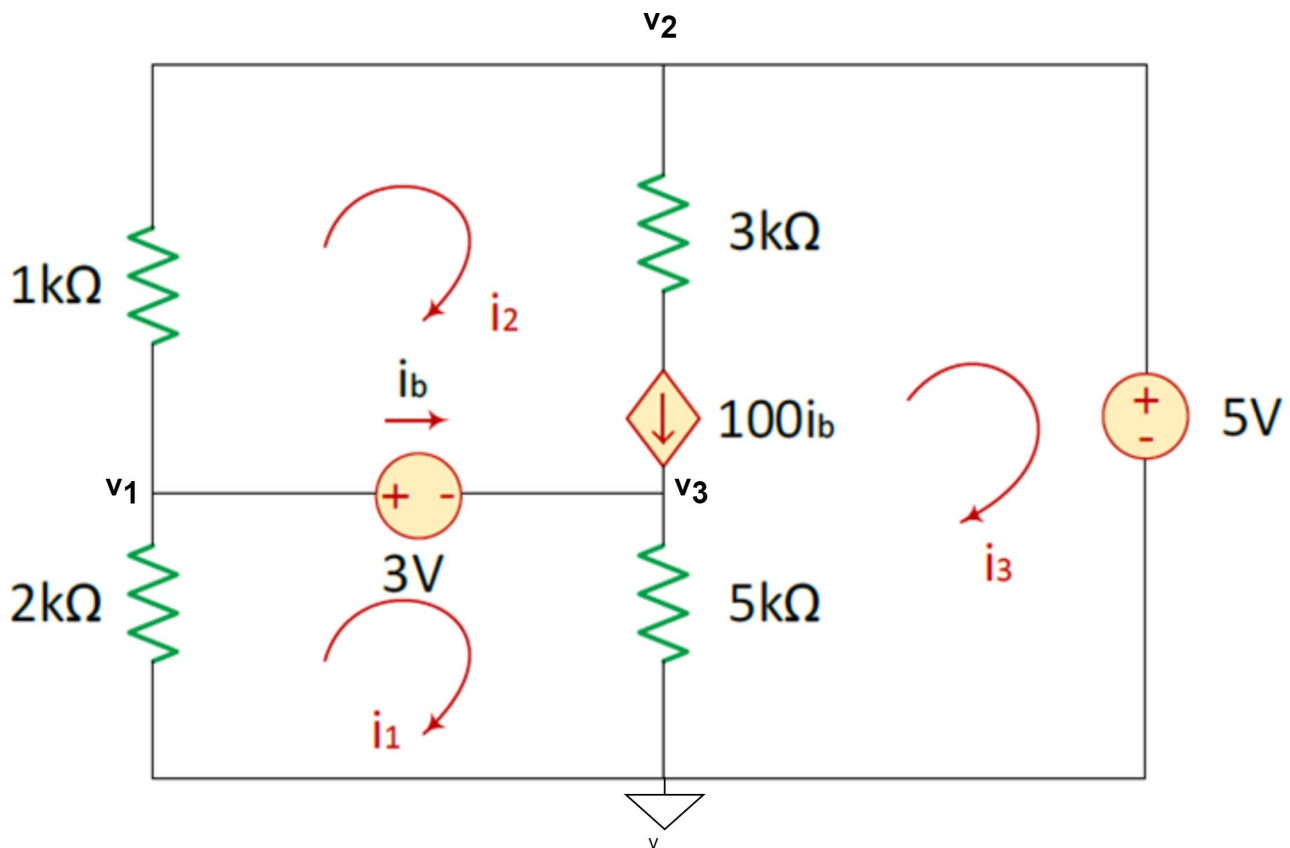
Transistors can serve as amplifiers, producing both current gain and voltage gain. For the transistor amplifier shown in Figure 3

i) Apply mesh analysis and write down the mesh equations using the labels provided.

Note: Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a, b, c and d can be positive or negative numbers, but DO NOT solve them.

ii) Apply nodal analysis and write down the nodal equations using the labels provided.

Note: Simplify the equations in the form $ev_1 + fv_2 + gv_3 = h$, where coefficients e, f, g , and h can be positive or negative numbers, but DO NOT solve them.



Answers:

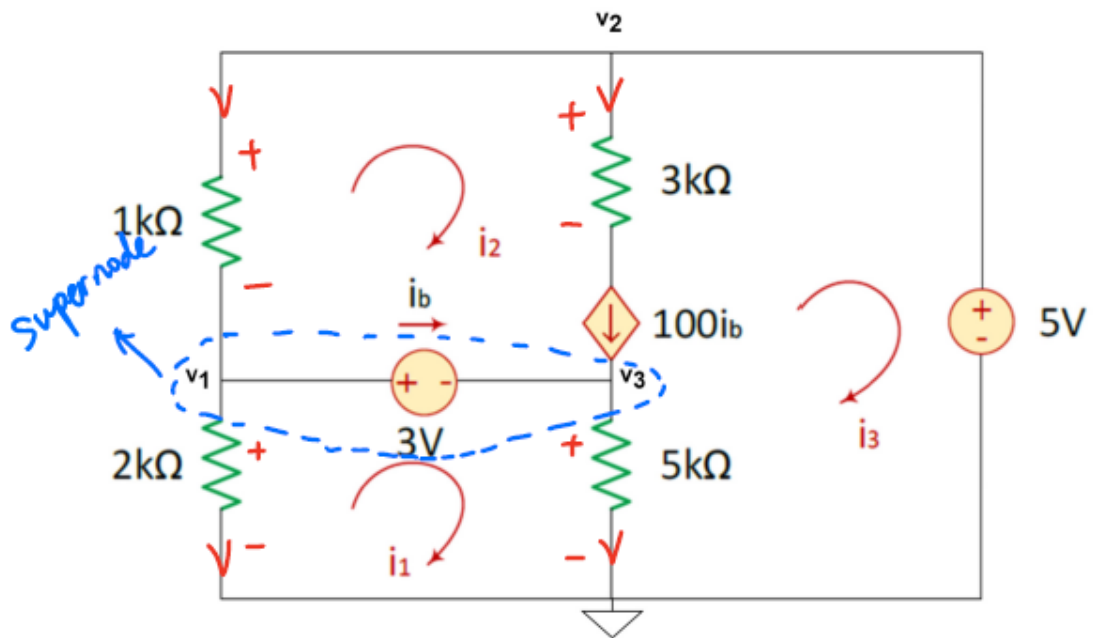
i) Mesh equations

$$\begin{cases} 7000i_1 - 5000i_3 = -3 \\ -5000i_1 + 1000i_2 + 5000i_3 = -2 \\ -100i_1 + 101i_2 - i_3 = 0 \end{cases}$$

ii) Nodal equations

$$\begin{cases} -1515v_1 + 1010v_2 - 2v_3 = 0 \\ v_2 = 5 \\ v_1 - v_3 = 3 \end{cases}$$

Solution to the modified question (Q4 - ii):



KCL @ the supernode:

$$\frac{V_2 - V_1}{1000} + 100i_b = \frac{V_1 - 0}{2000} + \frac{V_3 - 0}{5000} \quad (1)$$

constraint equation: $V_1 - V_3 = 3V$ (2) ✓

$V_2 = 5V$ (3) ✓

$$\frac{V_2 - V_1}{1000} = i_b + \frac{V_1}{2000} \quad (4)$$

simplify the equations ①-④:

$$\begin{cases} 10(V_2 - V_1) + 10^6 i_b = 5V_1 + 2V_3 & \text{⑤} \end{cases}$$

$$\begin{cases} i_b = \frac{V_2 - V_1}{1000} - \frac{V_1}{2000} & \text{⑥} \end{cases}$$

⑥ → ⑤:

$$10V_2 - 10V_1 + 1000(V_2 - V_1) - 500V_1 = 5V_1 + 2V_3$$

$$-1515V_1 + 1010V_2 - 2V_3 = 0$$

* Simplified equations $\begin{cases} -1515V_1 + 1010V_2 - 2V_3 = 0 \\ V_2 = 5 \\ V_1 - V_3 = 3 \end{cases}$